SESSION 5: INTRODUCTION TO STATISTICS IN R

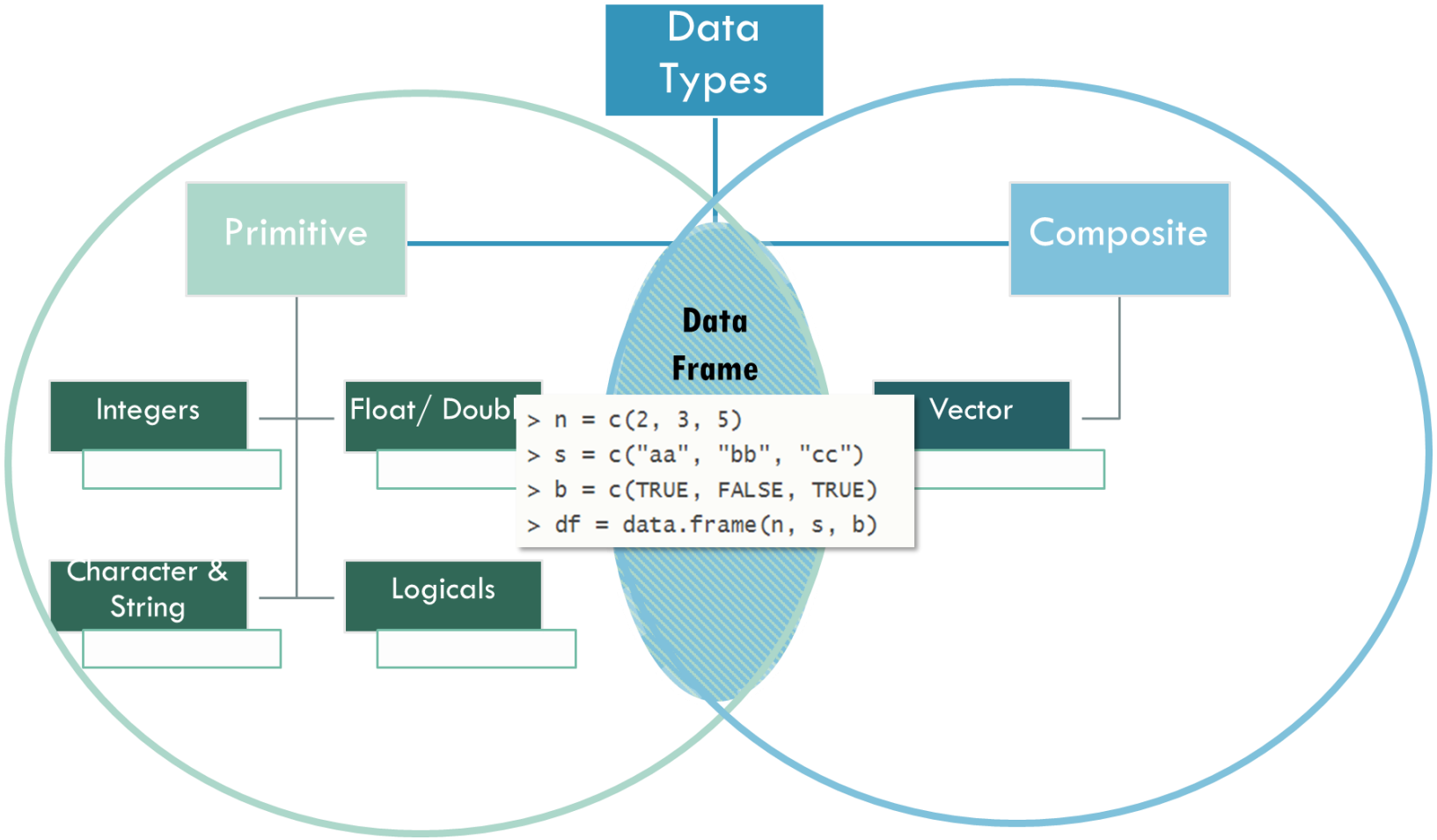
DR. SOFIA GIL-CLAVEL

- Recap
- Fundamentals of modeling in R: Example applied to linear models
- Other models?

1. SESSION 1-4: RECAP

- R and RStudio
- Data types
- Data frames

Introduction to R and Data frames



Introduction to R and Data frames

Using packages: Tidyverse

- Packages
- Tidy logic
- Dplyr

1.

```
> iris
# A tibble: 150 × 5
   Sepal.Length Sepal.Width Petal.Length Petal.Width Species
          <dbl>       <dbl>        <dbl>       <dbl> <fct>
 1          5.1         3.5          1.4         0.2 setosa
 2          4.9         3            1.4         0.2 setosa
 3          4.7         3.2          1.3         0.2 setosa
 4          4.6         3.1          1.5         0.2 setosa
 5          5           3.6          1.4         0.2 setosa
 6          5.4         3.9          1.7         0.4 setosa
 7          4.6         3.4          1.4         0.3 setosa
 8          5           3.4          1.5         0.2 setosa
 9          4.4         2.9          1.4         0.2 setosa
10          4.9         3.1          1.5         0.1 setosa
# ℹ 140 more rows
# ℹ Use `print(n = ...)` to see more rows
```

2.

Basic piping

- `x %>% f` is equivalent to `f(x)`
- `x %>% f(y)` is equivalent to `f(x, y)`
- `x %>% f %>% g %>% h` is equivalent to `h(g(f(x)))`

3.

dplyr is a grammar of data manipulation, providing a consistent set of verbs that help you solve the most common data manipulation challenges:

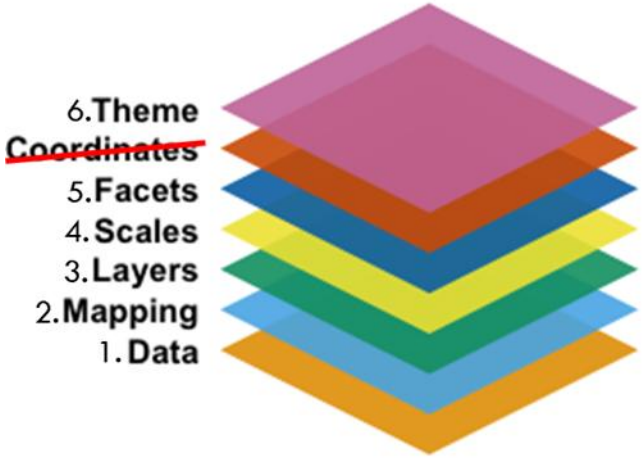
- `mutate()` adds new variables that are functions of existing variables
- `select()` picks variables based on their names.
- `filter()` picks cases based on their values.
- `summarise()` reduces multiple values down to a single summary.
- `arrange()` changes the ordering of the rows.

Introduction to R and Data frames

Using packages: Tidyverse

- Grammar of graphs
- Ggplot2
- Graphs components

Data Visualizations with ggplot2




```
ggplot(mpg, aes(x = displ, y = hwy,
                colour = class
                shape = drv
                size = cyl
                )) +
geom_point()
```

1.Data

2. Mapping: Variables in the Data

4. Scales

Add and transform the graph

3. Scale: Geometric Object

Introduction to R and Data frames

Using packages: Tidyverse

Data Visualizations with ggplot2

Unsupervised Learning

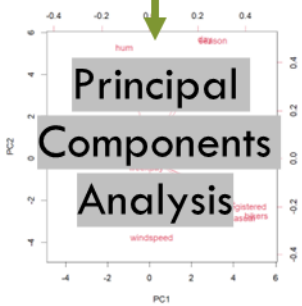
- 1.1 Statistical learning
- 1.2 Supervised vs. Unsupervised
- 1.3 PCA
- 1.4 K-means

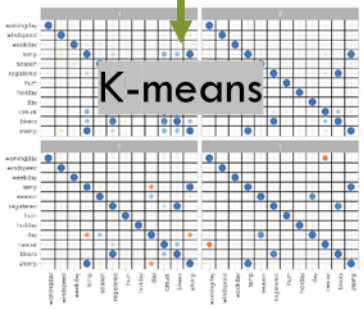
$$\cancel{Y_i} \sim X1_i + X2_i + X3_i$$

Numeric Matrix

Dimensionality reduction

Clustering





Introduction to R and Data frames

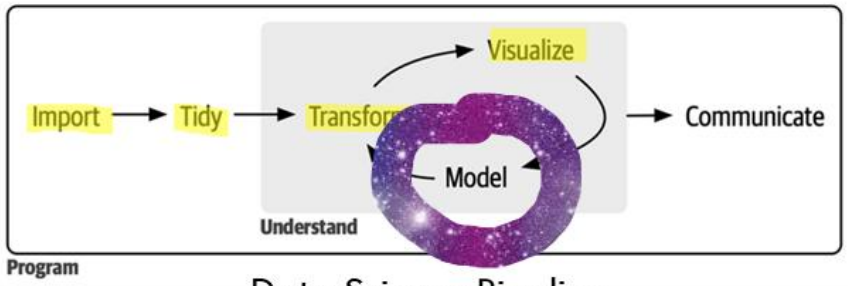
Using packages: Tidyverse

Data Visualizations with ggplot2

Unsupervised Learning

- Symbolic language
- Inference & prediction
- Fitting models
- Confidence intervals
- Statistical tests

Linear Models



Data Science Pipeline

stats-package {stats} R Documentation

The R Stats Package

Description

R statistical functions

Details

This package contains functions for statistical calculations and random number generation.

For a complete list of functions, use `library(help = "stats")`.

Author(s)

R Core Team and contributors worldwide

Maintainer: R Core Team R-core@r-project.org

We are entering the middle of the course!

Data Handling

1. Data Frames
2. Tidyverse

Exploratory Data Analysis

3. DataViz
4. Unsupervised Learning

Machine Learning (ML)

5. Statistics
6. Supervised Learning

Text Analysis

7. Text as Data
8. Text Mining
9. ML for Text

Code Optimization

10. Basic Programming
11. Servers & Parallelization

2. FUNDAMENTALS OF R-MODELING

TYPES OF VARIABLES

- **Dependent**: The variable (also known as outcome) we are analyzing and that we believe its behavior "depends" on other variables.

$$Y \sim$$

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TYPES OF VARIABLES

Symbolic Language

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Symbolic Language

$$\underbrace{Y \sim X1 + X2 + X3}$$

R was built to understand symbolic language!

BUT FIRST, LET'S OPEN SOME DATA

For this section, we will use the database "**ChickWeight**" from the basic R package.

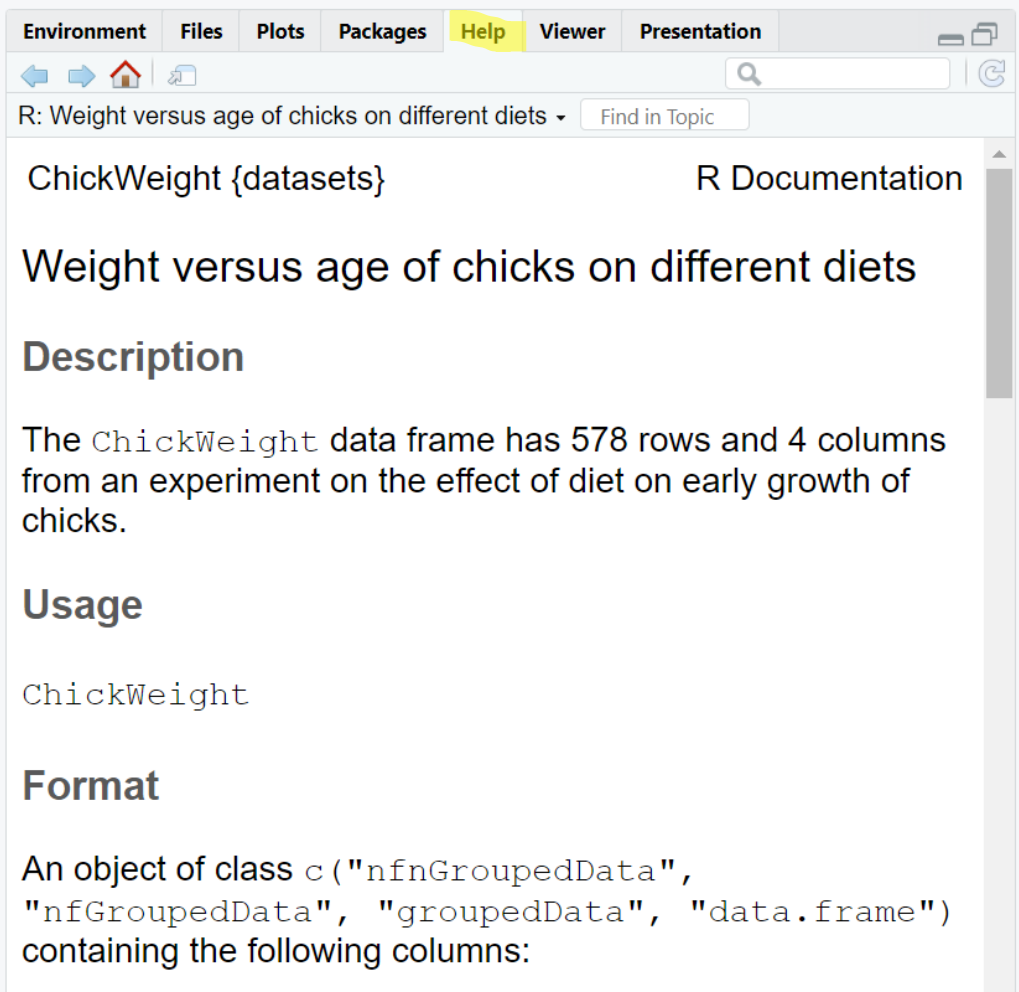
To open it, you just need to write down the name in the console:

```
ChickWeight
```

To see in an independent window, you can use:

```
View(ChickWeight)
```

BUT FIRST, LET'S OPEN SOME DATA



To learn what each variable represents, you can check its documentation in the "help" window.

BUT FIRST, LET'S OPEN SOME DATA

Environment | Files | Plots | Packages | Help | Viewer | Presentation

R: Weight versus age of chicks on different diets — Find in Topic

ChickWeight {datasets} R Documentation

Weight versus age of chicks on different diets

Description

The `ChickWeight` data frame has 578 rows and 4 columns from an experiment on the effect of diet on early growth of chicks.

Usage

```
ChickWeight
```

Format

An object of class `c("nfnGroupedData", "nfGroupedData", "groupedData", "data.frame")` containing the following columns:

Environment | Files | Plots | Packages | Help | Viewer | Presentation

R: Weight versus age of chicks on different diets — Find in Topic

weight

a numeric vector giving the body weight of the chick (gm).

Time

a numeric vector giving the number of days since birth when the measurement was made.

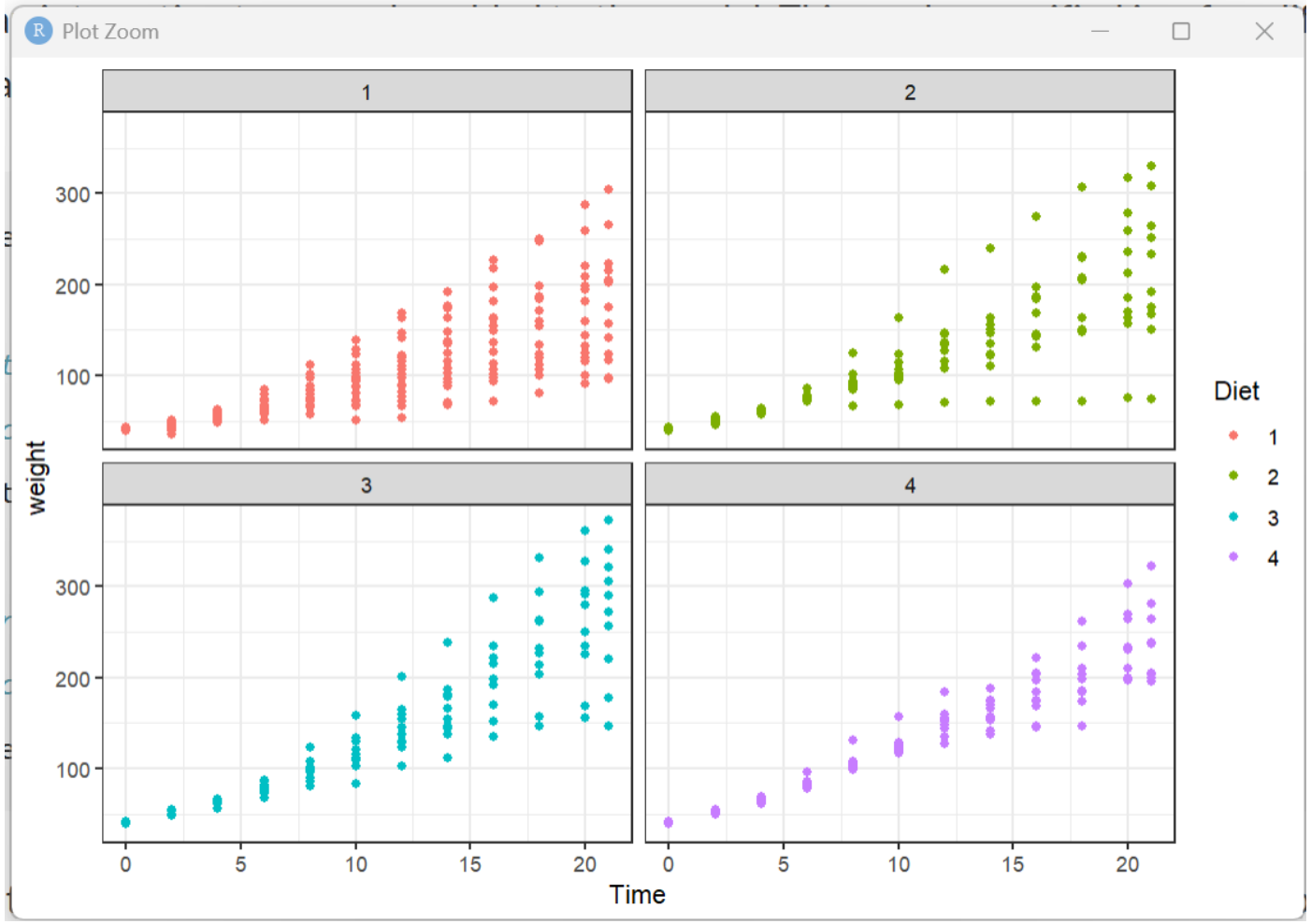
Chick

an ordered factor with levels `18` < ... < `48` giving a unique identifier for the chick. The ordering of the levels groups chicks on the same diet together and orders them according to their final weight (lightest to heaviest) within diet.

Diet

a factor with levels 1, ..., 4 indicating which experimental diet the chick received.

LET'S CHECK HOW THE DATA LOOKS



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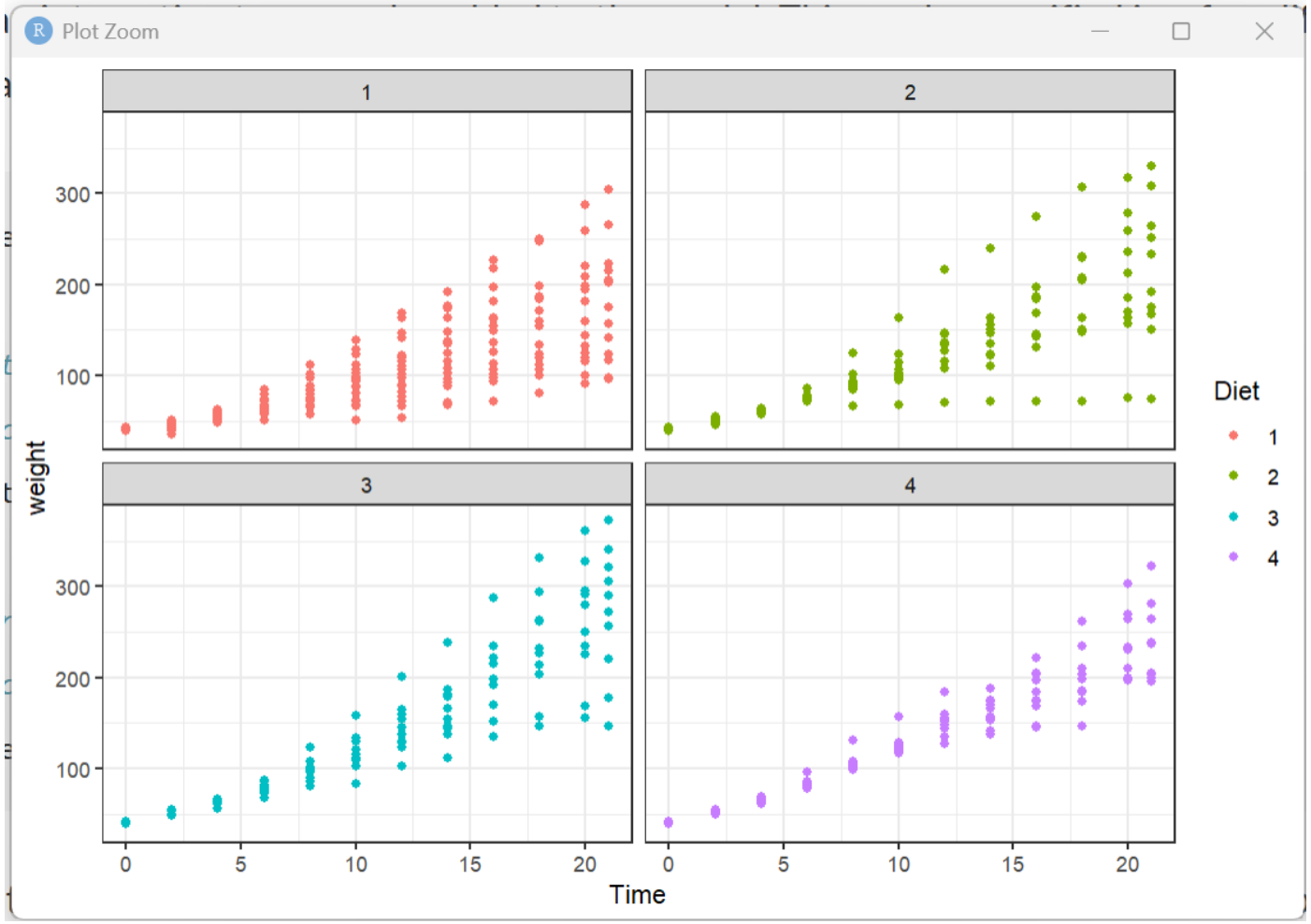

What are some interesting statistical questions?

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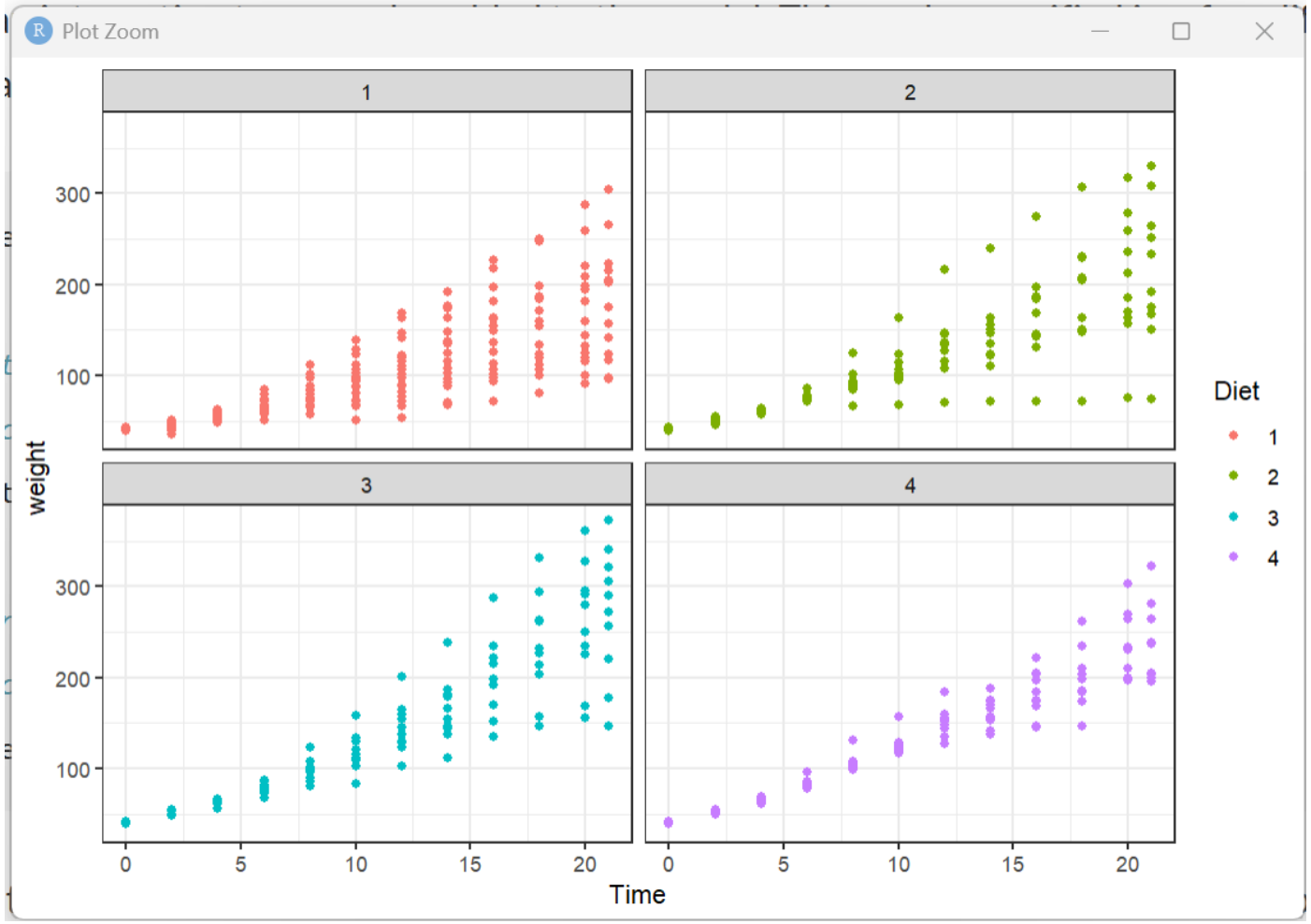

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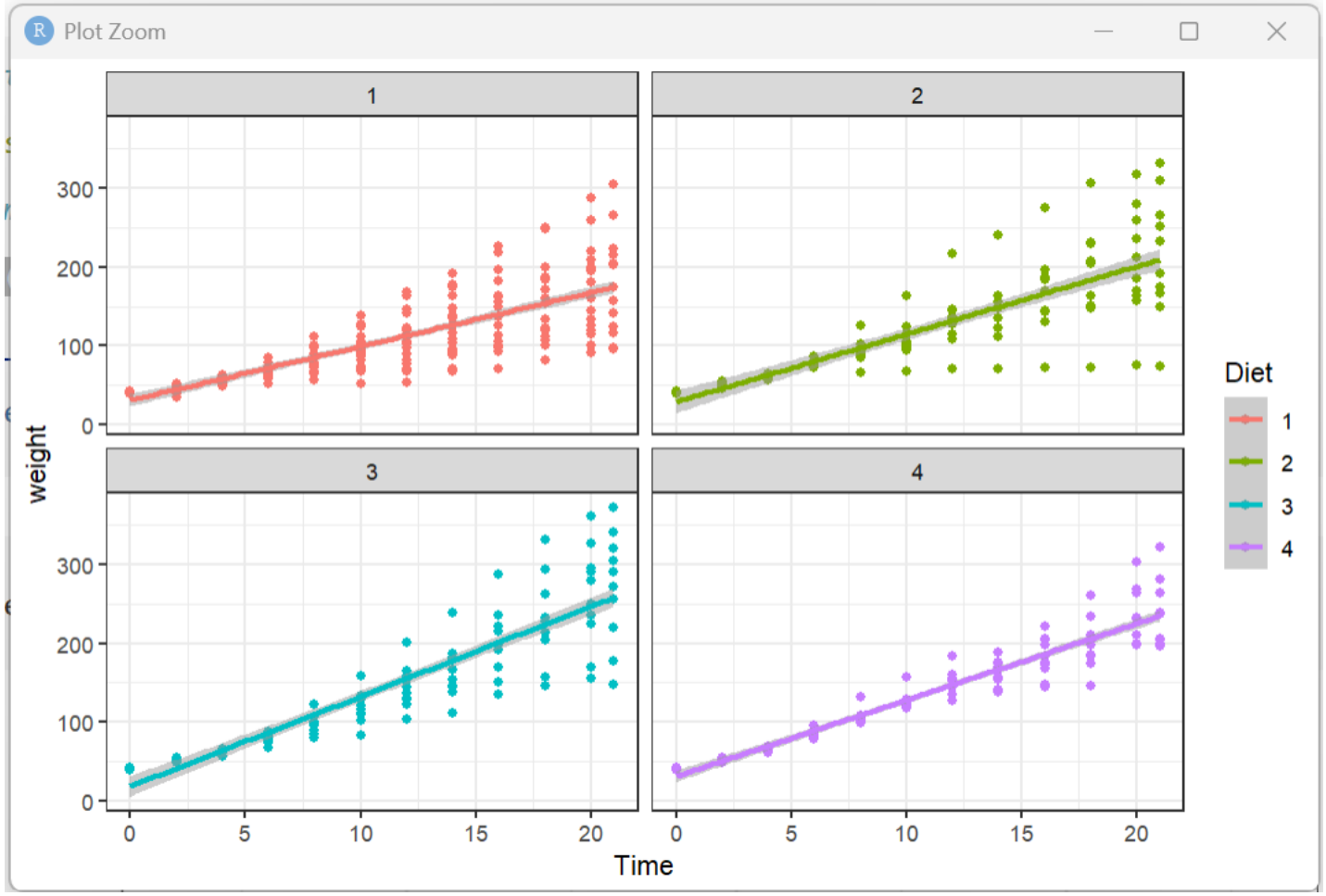

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Now we will learn about:

- Linear regression models

FITTING LINEAR MODELS

Environment | Files | Plots | Packages | Help | Viewer | Presentation

lm

R: Fitting Linear Models Find in Topic

lm {stats} R Documentation

Fitting Linear Models

Description

`lm` is used to fit linear models, including multivariate ones. It can be used to carry out regression, single stratum analysis of variance and analysis of covariance (although `aov` may provide a more convenient interface for these).

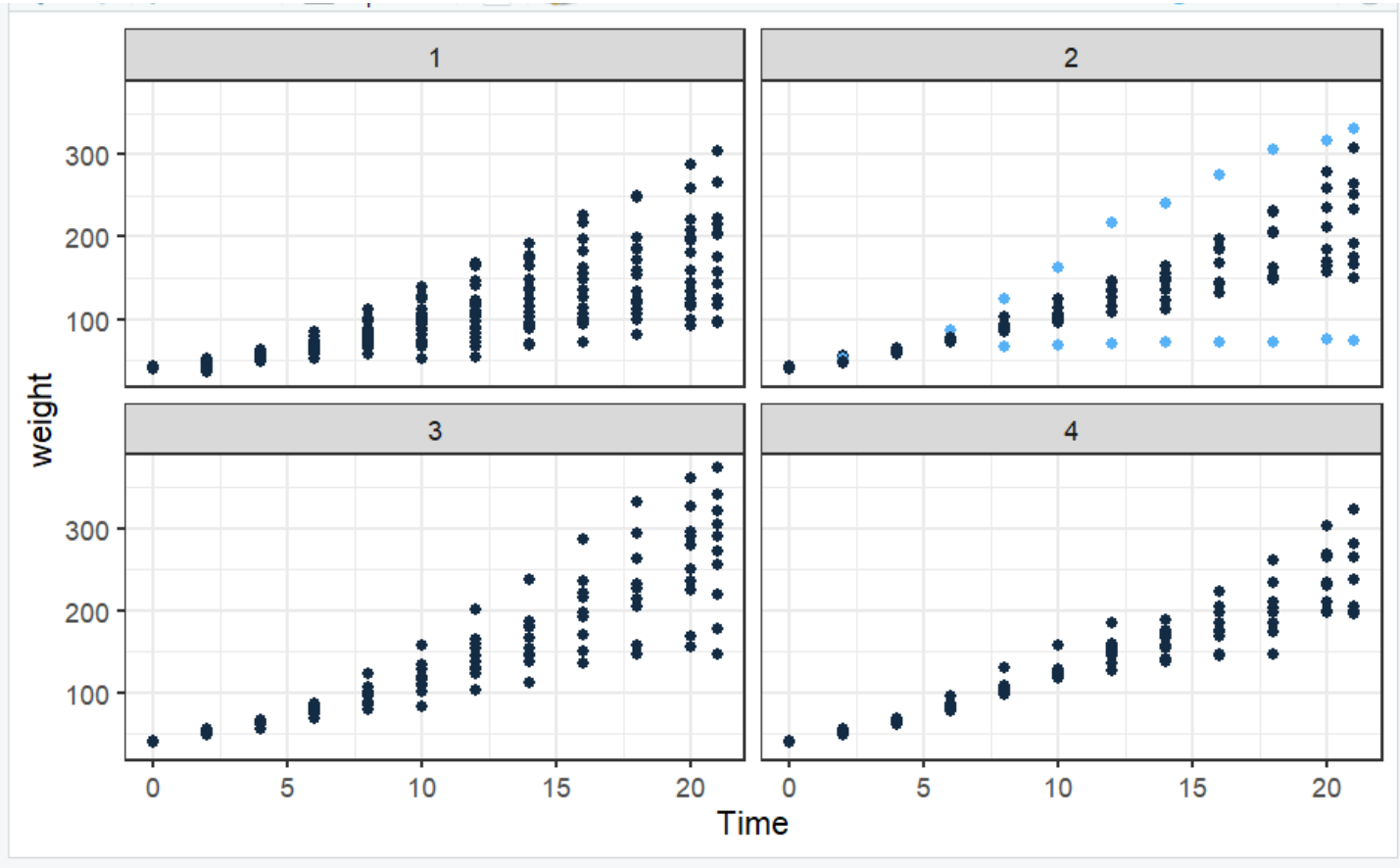
Usage

```
lm(formula, data, subset, weights, na.action,
   method = "qr", model = TRUE, x = FALSE, y = FALSE, qr = TRUE,
   singular.ok = TRUE, contrasts = NULL, offset, ...)
```

HANDLING OUTLIERS

An outlier can be higher or lower than expected or **displaced more to the right or left than expected.**

Outliers can affect regression lines, **making the regression lines less accurate** in predicting other data.



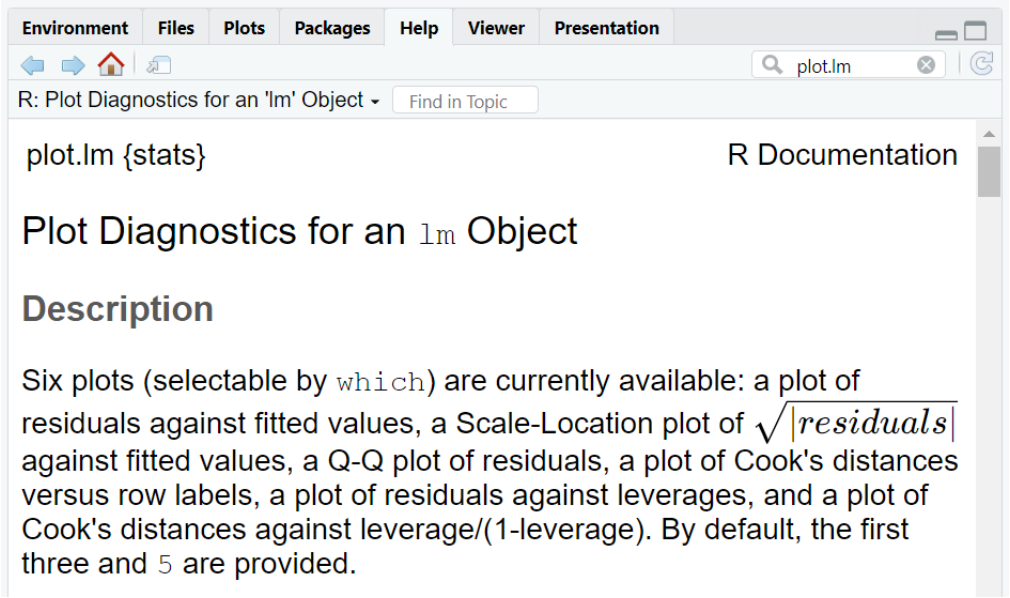
CHECKING THE ASSUMPTIONS

There are five key assumptions of linear regression:

1. 📈 **Linearity**
2. 🔵 **independence**
3. 📊 **homoscedasticity**
4. 🔔 **normality**
5. 🚫 **no multicollinearity**

Ensuring these assumptions are met is critical to creating an accurate and reliable model for predicting and drawing insights from data.

CHECKING THE ASSUMPTIONS



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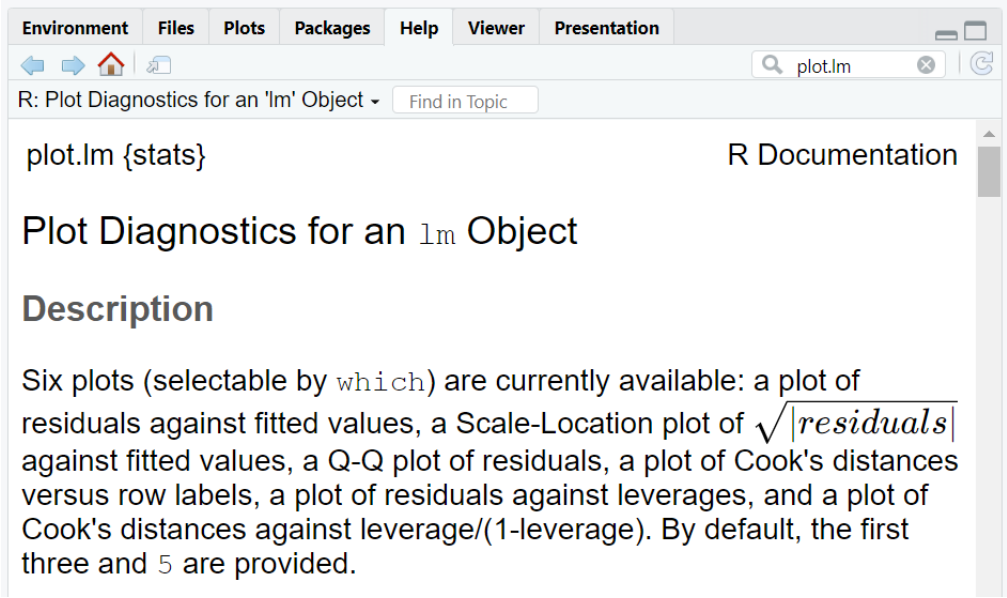
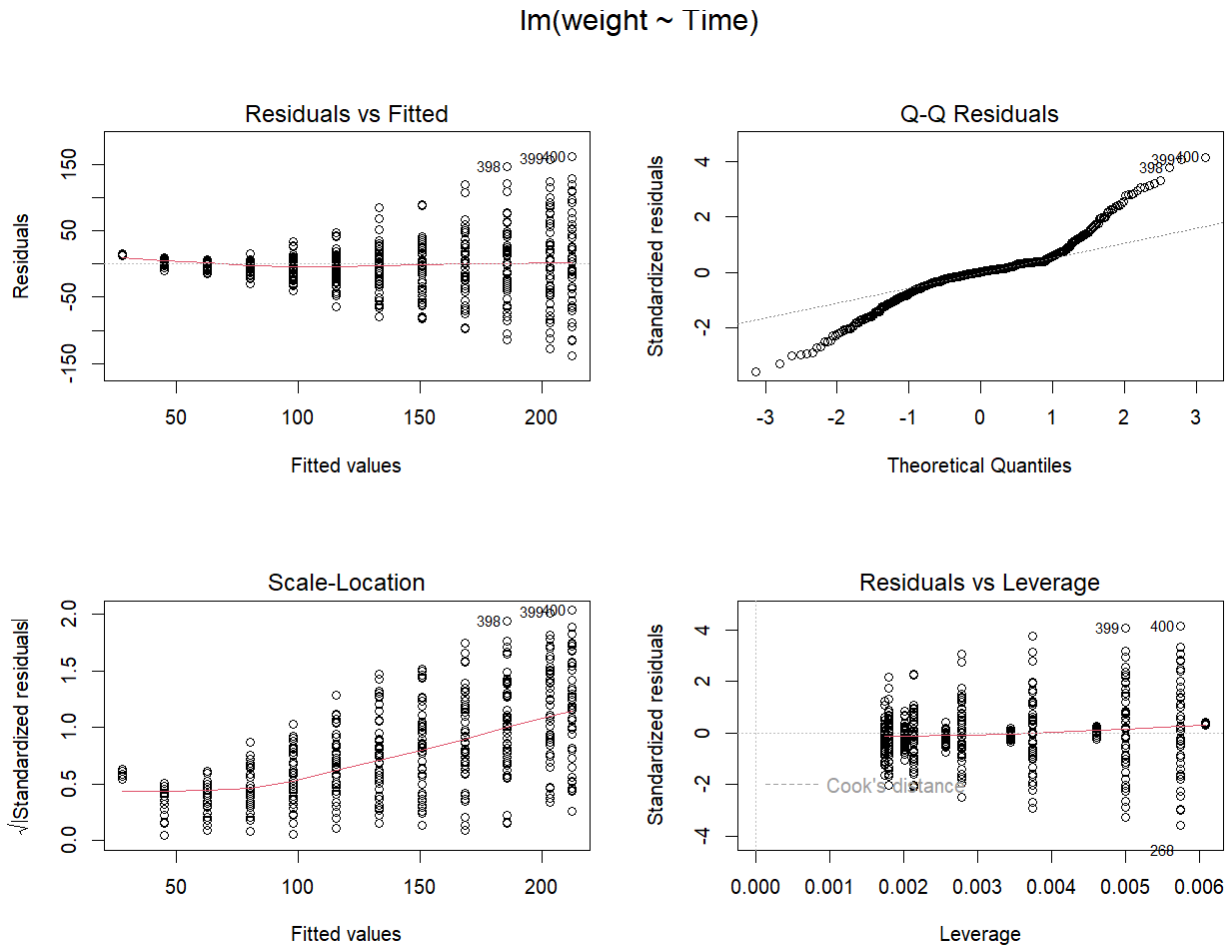
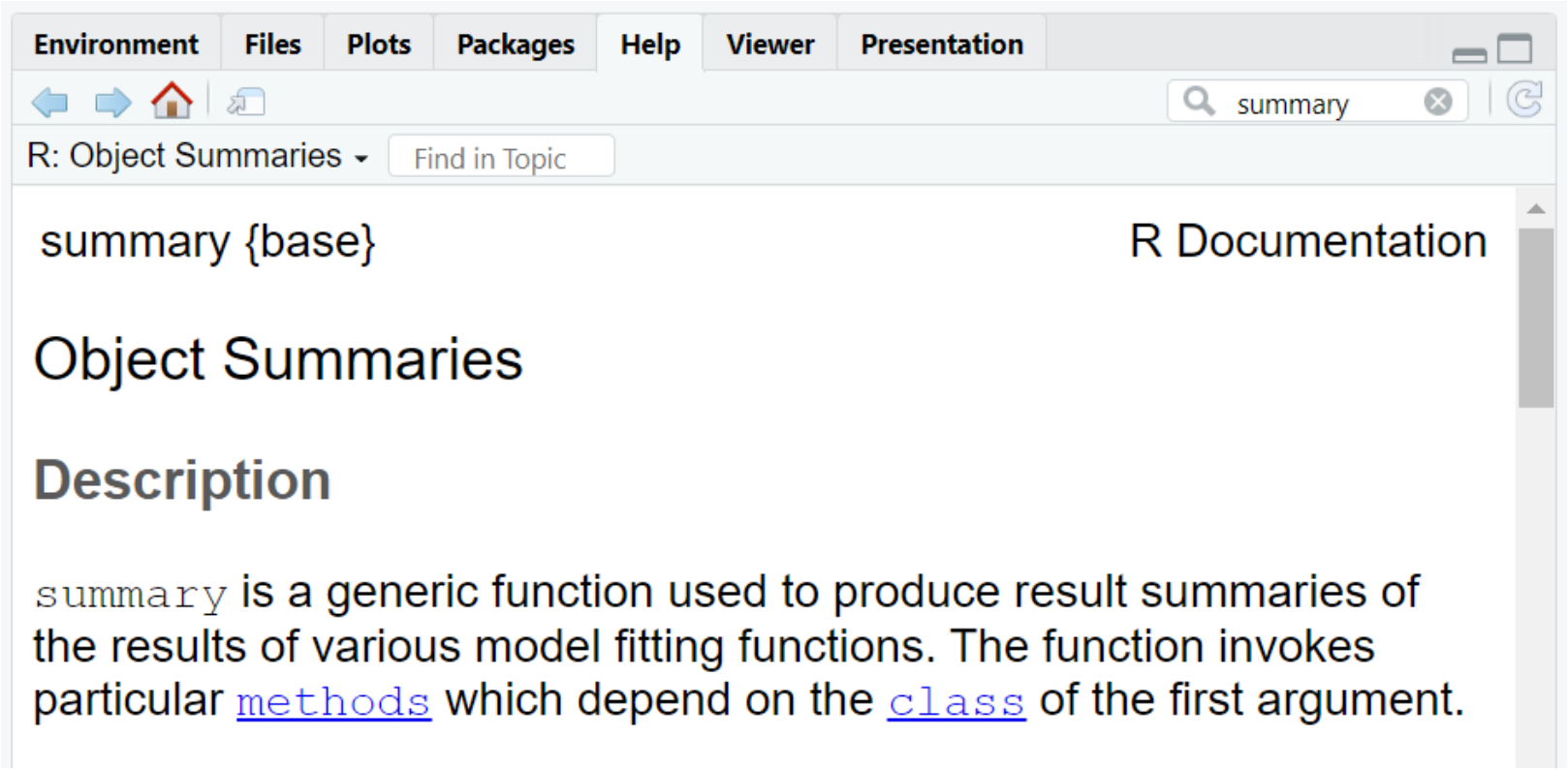
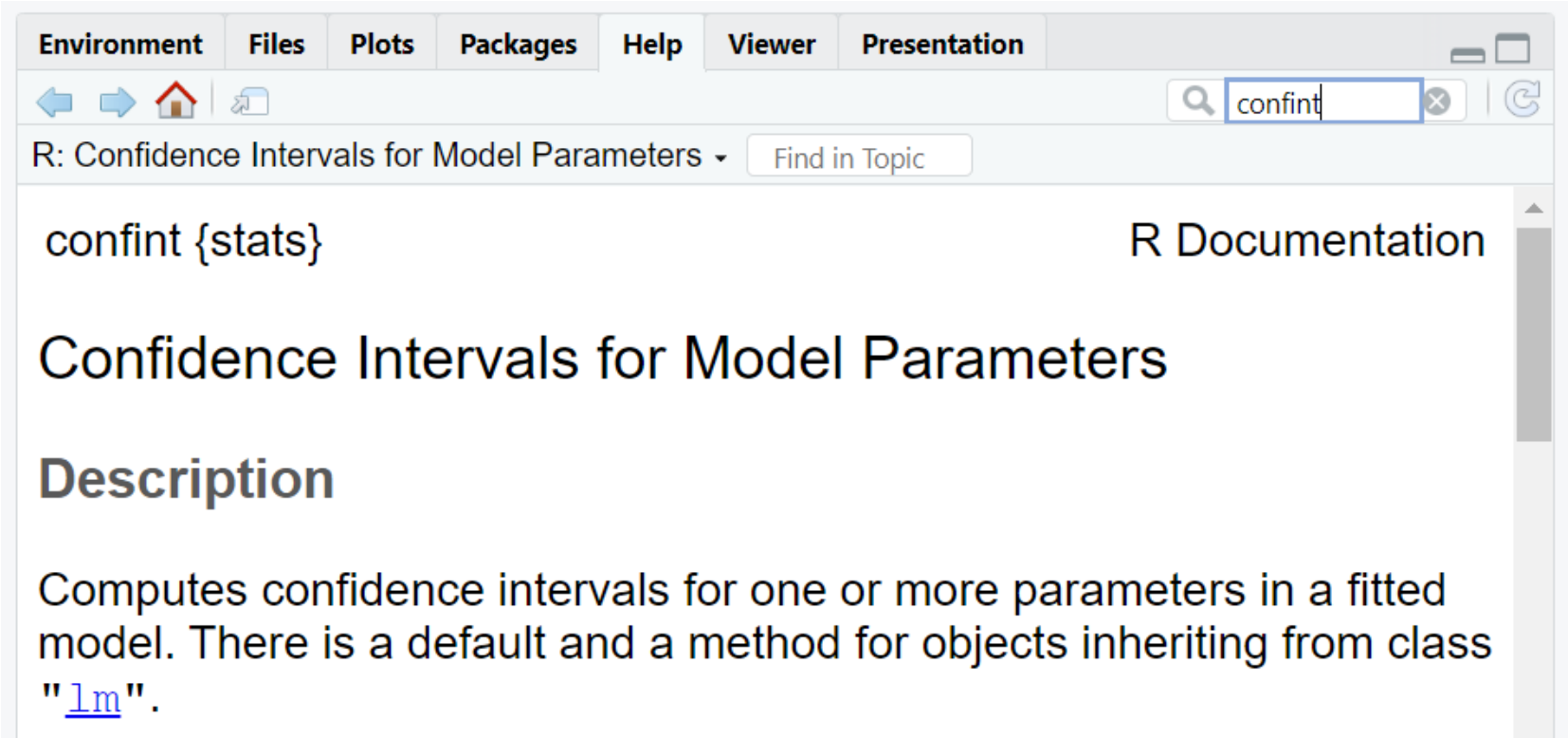




CHECK THE COEFFICIENTS AND OTHER STATS



CONFIDENCE INTERVALS



INTERACTION TERMS

Statistical **interaction** means **the effect of one independent variable**(s) **on the dependent variable depends on the value of another independent variable**(s).

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So, in our data, we could think that **the effect that time has on the chickens' weight depends on the diet**:

$$\boldsymbol{weight \sim Time \times Diet}$$

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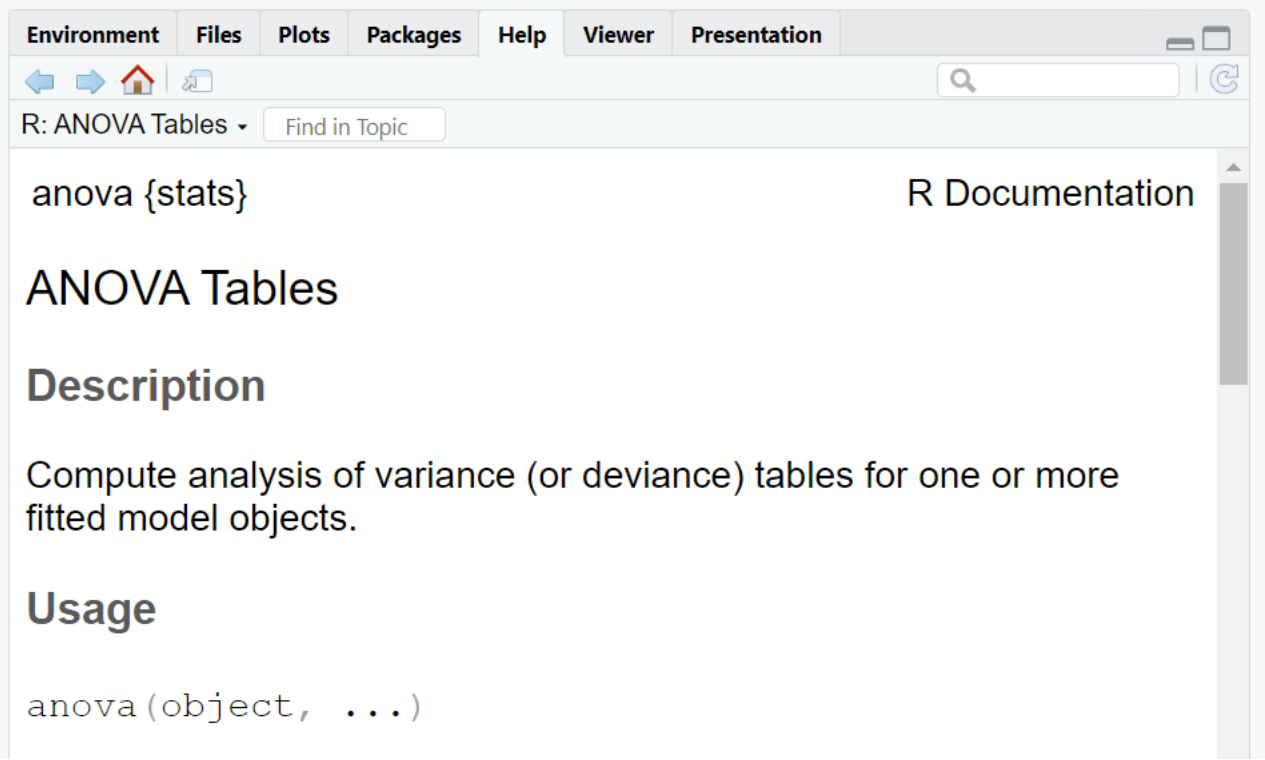
Remember, this is **Symbolic Language**

ANOVA: ANALYSIS OF VARIANCE

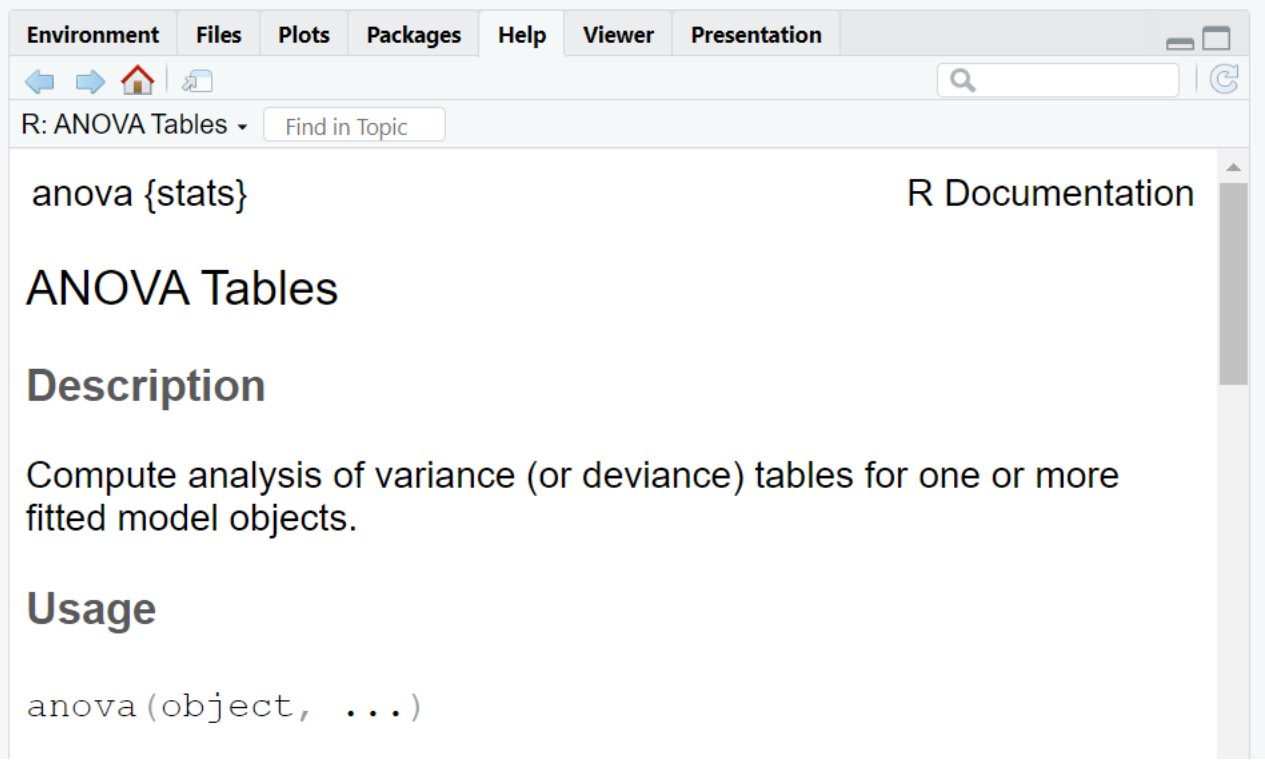
The ANOVA test is very useful when we want to compare two models to see which one fits the data better.



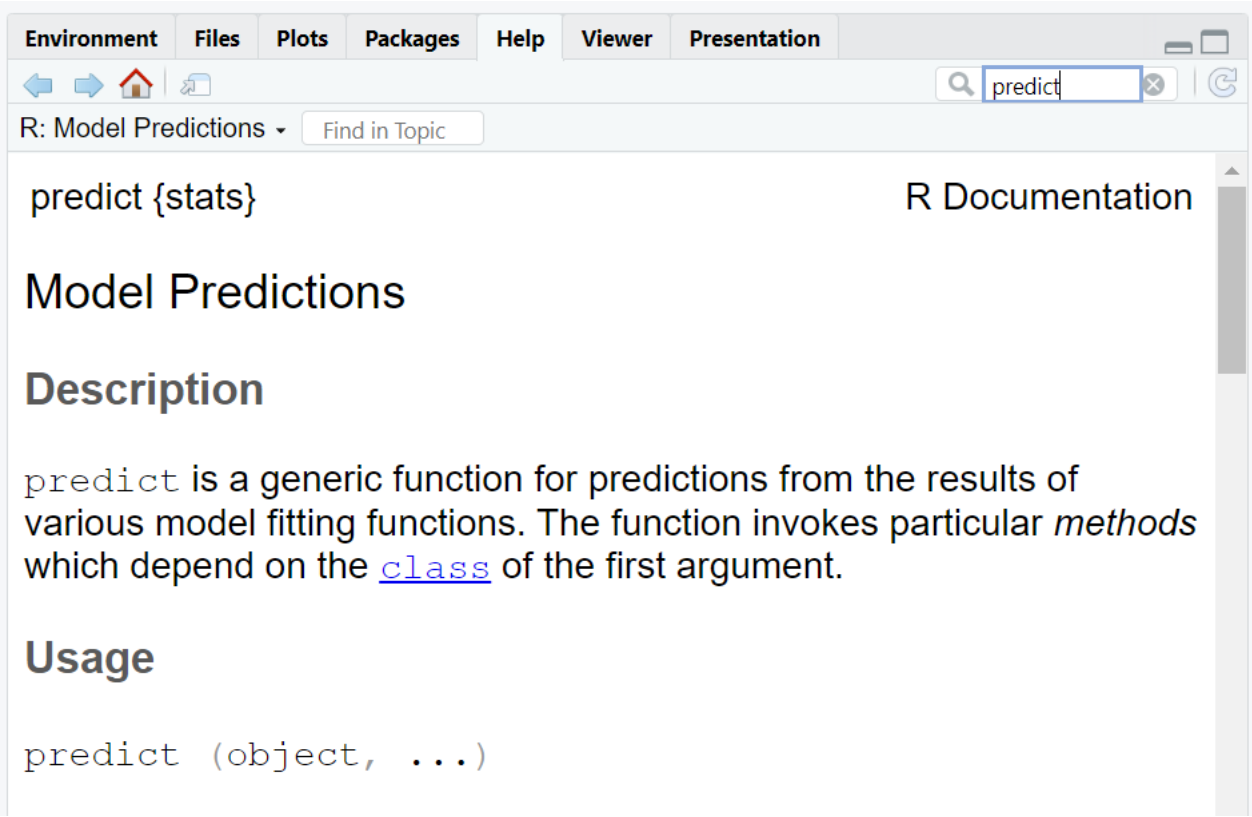
ANOVA: ANALYSIS OF VARIANCE

The ANOVA test is very useful when we want to compare two models to see which one fits the data better.

So, which model is better, with or without interaction term?



PREDICTING


Environment
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predict
R: Model Predictions
Find in Topic
predict {stats}
R Documentation
Model Predictions
Description
predict is a generic function for predictions from the results of various model fitting functions. The function invokes particular methods which depend on the class of the first argument.
Usage
predict (object, ...)

3. OTHER MODELS?

TYPES OF STATISTICAL ANALYSES AND MODELS

Based on the research question, we can perform:

- Descriptive
- Inferential
- Predictive
- Causal
- Etc….

Depending on the type of **Dependent** variable are the models we can use:

- Linear Model: Continuous
- Logit Model: Dichotomous
- Probit Model: Ordinal data
- Multinomial Model: Categorical Data
- Poisson/Binomial: Counting data

The relation between the **Dependent** and the **Independent** dictates the model structure:

- Nested model
- Bayesian

TYPES OF STATISTICAL ANALYSES AND MODELS

R has them all!

Based on
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TYPES OF STATISTICAL ANALYSES AND MODELS

Based on the question, we

- Descriptive
- Inferential
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- Etc….

R has them all!

They all work with the functions that we learned today. Sometimes, it is just a matter of adjusting some parameters.

- Multinomial Model: Categorical Data
- Poisson/Binomial: Counting data

HOMEWORK

OJ {ISLR2} R Documentation

Orange Juice Data

Description

The data contains 1070 purchases where the customer either purchased Citrus Hill or Minute Maid Orange Juice. A number of characteristics of the customer and product are recorded.

Send it by April 17th
In an email send me your commented r script

Using all your new knowledge, try to statistically analyze the OJ data from the package ISLR2 in R.

1. The dependent variable is “Purchase”. Turn this variable into dichotomous, i.e., it should take values of 0 and 1.
2. Perform some exploratory data analysis of the variables.
3. Perform a statistical analysis. For this use the function “glm()” with the parameter: family = "binomial". This means that “glm()” will be used as a logit model. You can check this post for more help: https://stats.oarc.ucla.edu/r/dae/logit-regression/

Join us!

https://forms.office.com/e/8Bgd2YsasJ

All about the lab:

https://societal-analytics.nl/

Contact us at:

analytics-lab.fsw@vu.nl.

https://sofiag1l.github.io/

THANKS!

Dr. Sofia Gil-Clavel